# Isoline

## Reads the ECG. Knows when not to.

Quality-aware, explainable ECG screening from scans and phone photos

Talha Salman (560) · Deep Learning Semester Project

# The problem

- ECGs often travel as printouts and phone photos, not as clean signals
- Real photos are tilted, creased, shadowed and blurry
- A network always gives an answer, sure or not
- A wrong answer that looks confident is the dangerous case

# The idea: five steps

- 1 Check the picture: good, fair or bad
- 2 Read the image with an EfficientNet
- 3 Rebuild the waveform with a digitizer
- 4 Cross-check with a signal model (CNN + Transformer)
- 5 Explain with a heat map and a one-page report

# Dataset: PTB-XL

**21,388**
ECG records with a diagnosis

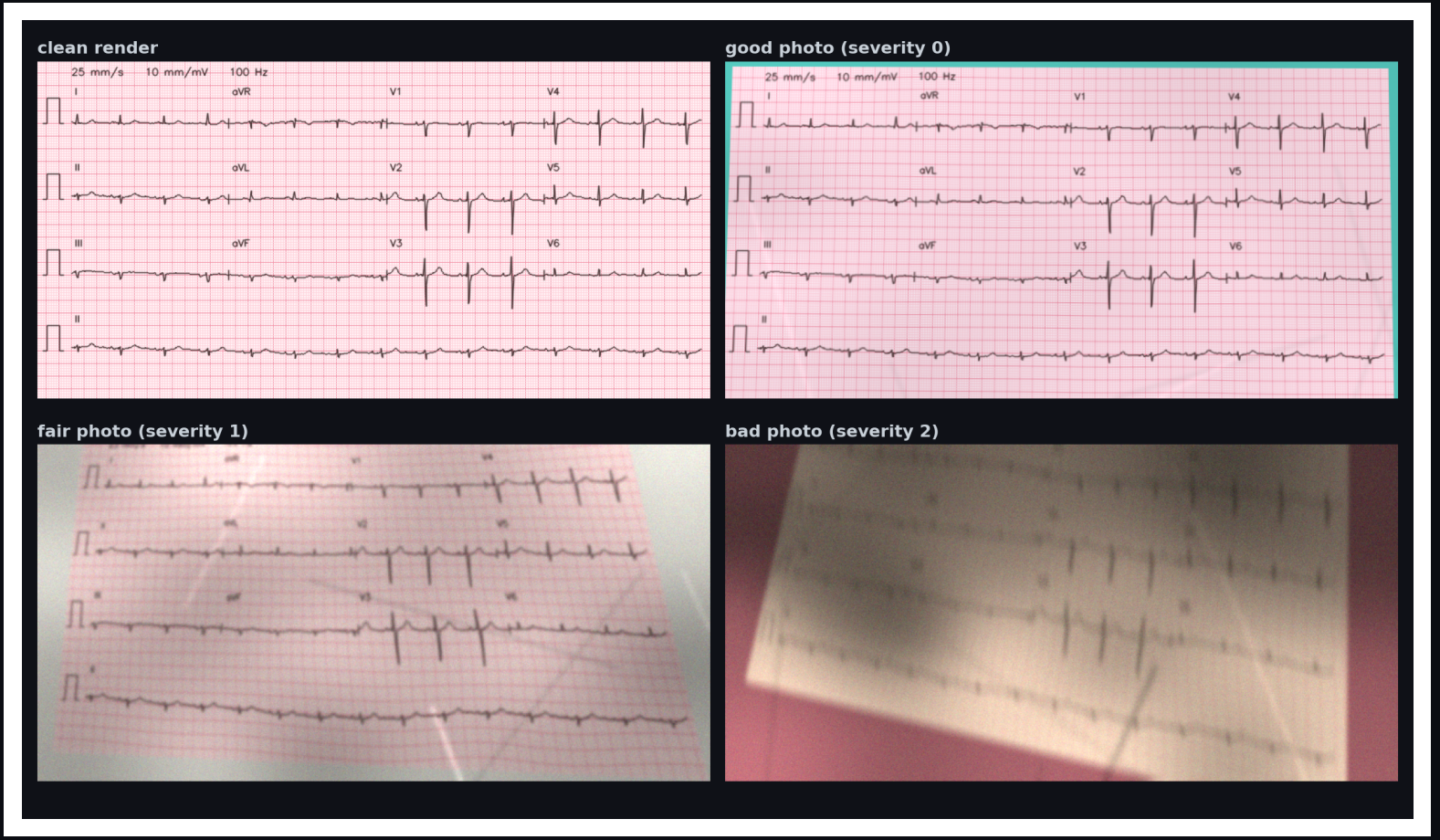
**5**
classes: NORM, MI, STTC, CD, HYP

**12**
leads, 10 seconds, 100 Hz

**42,776**
pictures made from them

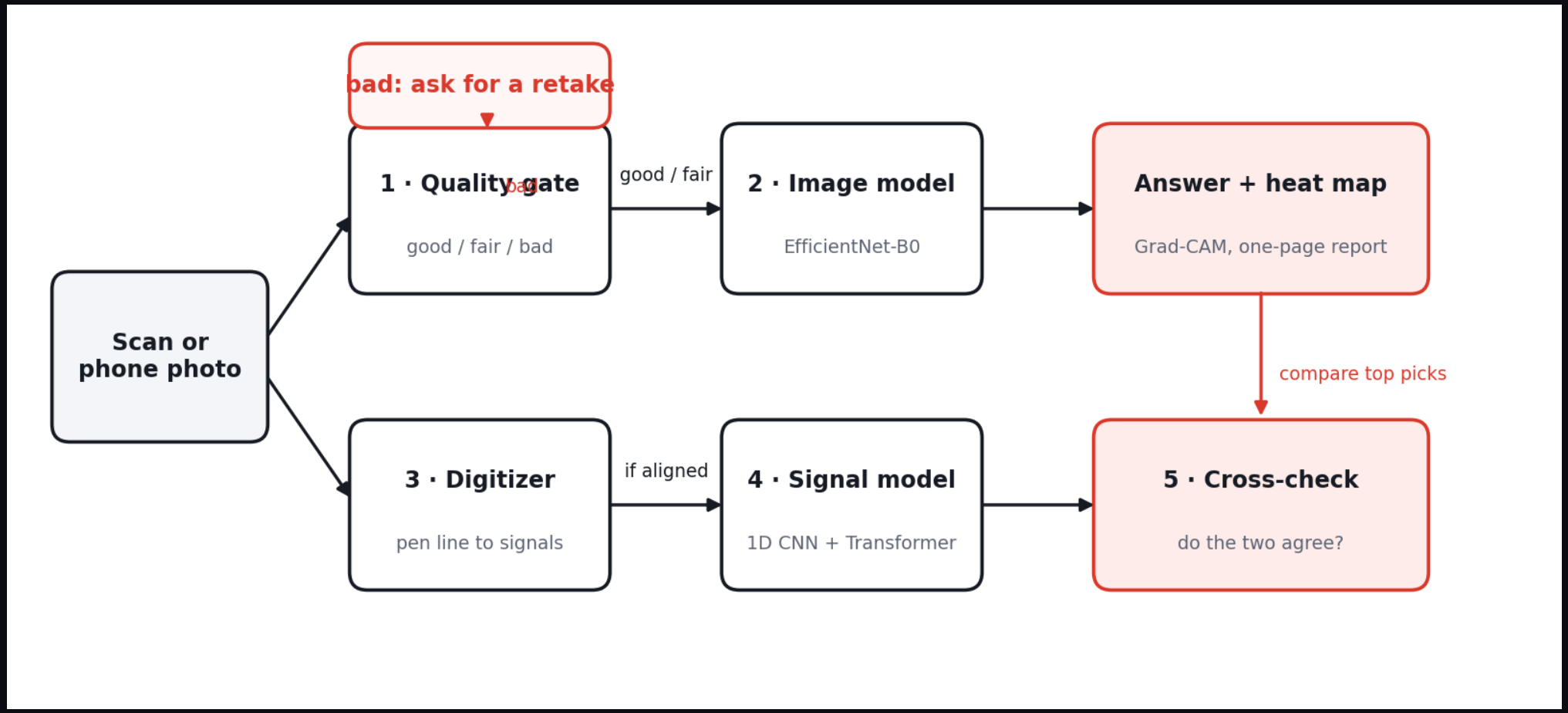
- Public, cardiologist-annotated, CC BY 4.0 (PhysioNet)
- Official split: folds 1 to 8 train, 9 validation, 10 test

# Making the pictures




One ECG: a clean print, then phone photos at three severities

# The pipeline




The image model gives the answer; the waveform path is an independent cross-check

# The two models

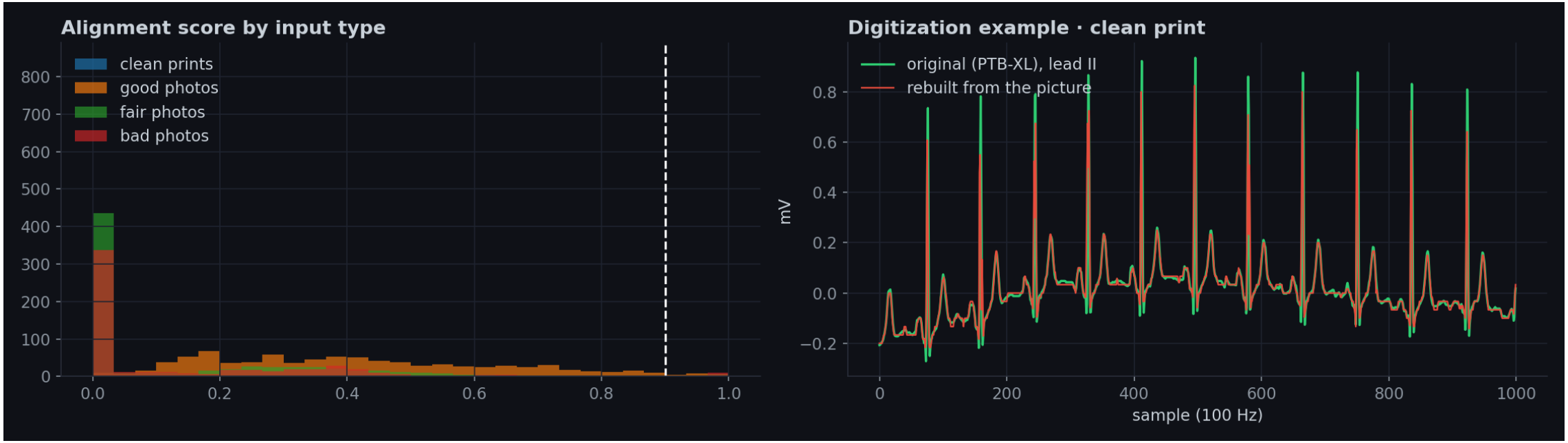
## Image model

- EfficientNet-B0, pre-trained on ImageNet
- Two heads: 5 diagnoses + photo quality
- Temperature-calibrated, thresholds tuned on validation
- Grad-CAM heat maps

## Signal model

- 1D residual CNN with squeeze-excitation
- 3-layer Transformer on top
- Fine-tuned on what a printout really shows (2.5 s per lead)
- Used as the cross-check

# Rebuilding the waveform




Alignment scores of clean prints and photos, and one rebuilt lead

# Results on the held-out test set

**0.929**
macro AUC, signal model

**0.921**
macro AUC, image model

**99.1%**
photo-quality gate accuracy

**0.926**
signal model, printout layout

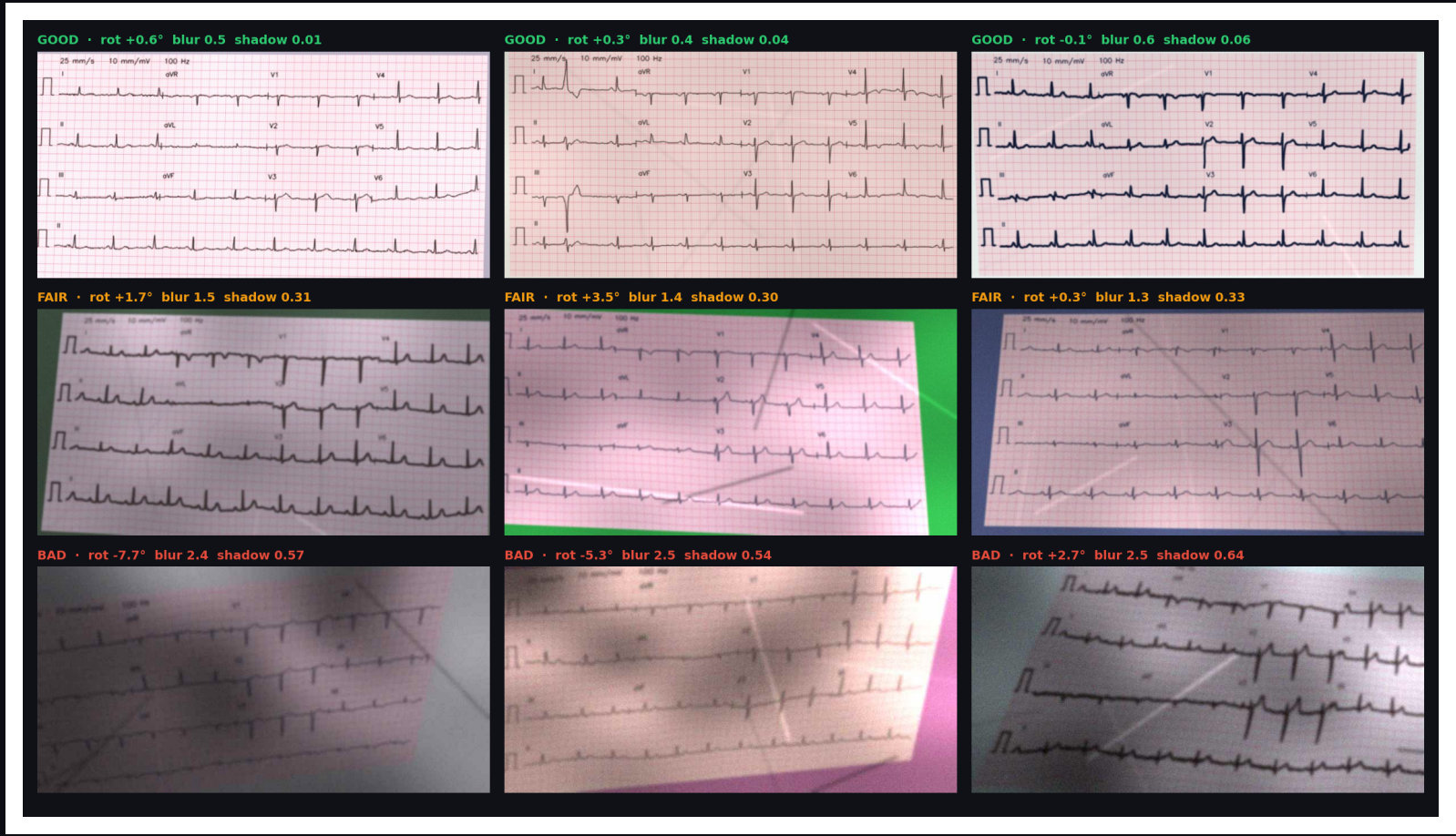
- AUC averaged over five classes; hypertrophy is the hardest
- Rebuilding the signal from a picture costs accuracy (about 0.814 AUC), so the image model answers

# Answering less, but more reliably

| When Isoline answers | Answered | Top-1 accuracy |
|---|---|---|
| A · answer every image | 100.0% | 82.4% |
| B · answer unless the quality gate rejects | 87.6% | 83.0% |
| C · B, but withhold when the two models disagree | 59.8% | 84.9% |
| **D · answer only when both models agree** | **22.8%** | **89.6%** |

Top-1 counts a prediction as right if it matches any recorded label of the patient.

# Explaining each answer




What the image model sees, with its output

# Built to be run by anyone

- GitHub: code, trained weights, notebook, example pictures
- One command: python predict.py --image samples/ecg_01.jpg
- Website on Vercel: dataset, results, real example outputs, report, slides, video
- Live demo on Hugging Face Spaces

# Limits, said plainly

- Not a medical device: a learning project and a screening aid
- Trained on rendered ECGs, not tested on real clinic photos
- Strongly tilted photos cannot be cross-checked yet
- Fewer than one pixel per sample: sharp peaks are slightly clipped

# Thank you

## Reads the ECG. Knows when not to.

https://github.com/talhahSalman/isoline-ecg
Questions welcome